

Meraf Hailesslassie

Minneapolis, MN | merafhailesslassie@gmail.com | 507-581-6918 | meraf255.github.io

Research interests: applying well-established quantitative methods — spatial statistics, causal inference, time-series forecasting — in service of specific public health questions, to build evidence for policy on chronic disease and maternal & child health, domestically and in low- and middle-income countries.

EDUCATION

St. Olaf College — Northfield, MN

Feb 2021 – Dec 2024

B.A., Mathematics; Quantitative Economics; Data Science and Statistics

Relevant coursework: Real Analysis I, Probability Theory, Modern Computational Math, Econometrics: Time Series and Forecasting, Statistical Modeling, Introduction to Data Science, Algorithms for Decision Making, Health Care Economics, Advanced Statistical Modeling

RESEARCH EXPERIENCE

Spatial Statistics Research Assistant, CURI

Jun 2023 – Dec 2023

St. Olaf College, Statistics Department — with Prof. Laura Boehm Vock

- Adapted and synthesized spatial statistical methods from multiple prior studies to estimate total volume change of Storglaciären, an internationally studied glacier in northern Sweden.
- Estimated and propagated error through Digital Elevation Models (DEMs) to quantify accuracy, precision, and uncertainty in volume-change calculations.
- Co-authored and presented an academic paper on the spatial statistical analysis of glacier volume loss as part of a research team (with Sonam Gurung, Prasiddha Shrestha, and Cheng Vang).

Access to Healthcare Services among Medicaid Beneficiaries

2023 – 2024

Independent research, St. Olaf College — with Wonu Falae

- Modeled the relationship between ethnicity, gender, and access to healthcare among Medicaid beneficiaries using multiple linear regression, logistic regression, stepwise selection, and lasso/ridge regression with cross-validation.
- Identified health status, program enrollment, and utilization — not ethnicity or gender — as the strongest predictors of access to care.

Research Assistant, Healthcare Economics

Spring 2023

St. Olaf College

- Developed an inpatient and outpatient Specialty Intensity Score to support evaluation of care coordination across providers.
- Processed Limited Data Set (LDS) Medicare claims data in R and identified key variables to construct a provider-diagnosis network for care-coordination analysis.

Research, Time Series and Forecasting

Fall 2023

St. Olaf College

- Investigated drivers of rising U.S. healthcare costs, including high prices, overtesting, and revenue-focused incentives.
- Built a structural VAR (vector autoregression) model to quantify the relationship between personal consumption factors (tobacco, insurance, housing/utilities) and healthcare spending trends, using annual BEA data from 1959–2022.

A Dynamic Model of Conflict & Resolution Strategies in Refugee Camps

2024

Differential Equations course project, St. Olaf College — with Aya Elawad

- Adapted the Lotka-Volterra predator-prey model to compare negotiation, mediation, and arbitration as conflict-resolution strategies in refugee camps, focused on the South Sudanese and Ethiopian refugee crises.
- Used equilibrium, nullcline, and Jacobian eigenvalue analysis to assess model stability under low-, moderate-, and high-conflict initial conditions.

Bay County Box Turtle Project: Ecological Recovery After Hurricane Michael

2024

Independent analysis, using Bay County, FL citizen-science data

- Built a multilevel regression model to account for repeated measurements on the same tagged turtles, examining how years since the hurricane, air temperature, and movement stationarity relate to turtle shell length as a proxy for health.

Research, Behavioral Economics

Spring 2024

St. Olaf College — with Lexie Ligday

- Applied game theory and behavioral economic principles to model food-waste reduction strategies for cafeteria sustainability initiatives, framing the problem as a principal-agent game between a catering company and student body.

Substrate Specificity of Yeast Alcohol Dehydrogenase

Spring 2023

Math 236: Mathematics of Biology, St. Olaf College — with Will Asinger and Calvin Reyes

- Investigated yeast alcohol dehydrogenase (ADH) activity across four straight-chain alcohols using the Michaelis-Menten model fit by nonlinear regression, comparing binding affinity (K_m) and enzyme turnover (V_{max}).

PROFESSIONAL EXPERIENCE

Business Intelligence Report Developer

May 2025 – Present

American Dairy Queen Corporation — Bloomington, MN

- Streamline audit-combination processes using Power BI, Databricks, PowerShell, and Excel VBA to unify and optimize data workflows across finance, R&D, marketing, and operations datasets.
- Automate previously manual reporting processes, improving accuracy, efficiency, and productivity.
- Contribute to the Inclusive Culture Council, helping sustain DEI initiatives and foster an inclusive workplace culture.

Data Science Intern

Jan 2025 – May 2025

Darcy Solutions Inc — Minneapolis, MN

- Designed and implemented an alert system to evaluate heat exchanger efficiency across multiple sites.
- Developed a scalable Power Apps solution to track well-building projects from initial client request through project completion.

Global Data Insight and Analytics Intern

Jun 2024 – Aug 2024

Ford Motor Company — Dearborn, MI

- Built and modified scripts to process supplier performance data, informing operational strategy across plants.
- Applied feature engineering techniques to forecast and mitigate premium freight events, reducing costs and improving supply chain resilience.

Labor Productivity and Benchmarking Intern

Jun 2022 – Aug 2022

Allina Health — Minneapolis, MN

- Analyzed quarterly productivity reports using Vizient and Workday to help managers benchmark hospital productivity.
- Supported implementation and training for PowerLabor, a system-wide productivity measurement application.

SKILLS

Statistical & Quantitative Methods: Spatial statistics, time series / VAR modeling, causal inference, multilevel modeling, regression & logistic regression, LDA / QDA, KNN, lasso/ridge regression, cross-validation, error propagation, Michaelis-Menten / nonlinear regression, game theory, feature engineering

Technical: R, Python, SQL, Power BI, Databricks, Alteryx, QlikSense, Excel (VBA), Power Apps, SharePoint, R Shiny

Languages: English (fluent), Amharic (fluent)